1 Angle update

$$
{}^{V[k]}\bar{\mathbf{A}}[k+1] \approx \begin{bmatrix} \hat{a}_{11}[k+1] & 0 & \hat{a}_{13}[k+1] \\ 0 & \hat{a}_{11}[k+1] & \hat{a}_{23}[k+1] \\ \hat{a}_{13}[k+1] & \hat{a}_{23}[k+1] & \hat{a}_{33}[k+1] \end{bmatrix}
$$

$$
\delta\hat{\theta}_u[k] = \frac{\hat{a}_{23}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]}
$$

$$
\delta\hat{\theta}_v[k] = -\frac{\hat{a}_{13}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]}
$$

$$
\delta\hat{\theta}_w[k] = -\tan\hat{\theta}_x[k]\,\delta\hat{\theta}_v[k]
$$

$$
\hat{\theta}_x[k+1] = \hat{\theta}_x[k] + \delta\hat{\theta}_u[k]
$$

$$
\hat{\theta}_y[k+1] = \hat{\theta}_y[k] + \frac{\delta\hat{\theta}_v[k]}{\cos\hat{\theta}_x[k]}
$$

2 Process in the local coordinate

$$
{}^{v}\bar{\mathbf{A}} = \hat{\mathbf{V}}^{T}\,\bar{\mathbf{A}}\,\hat{\mathbf{V}} = \begin{bmatrix} \bar{a}_{11} & 0 & \bar{a}_{13} \\ 0 & \bar{a}_{11} & \bar{a}_{23} \\ \bar{a}_{13} & \bar{a}_{23} & \bar{a}_{33} \end{bmatrix}
$$

$$
\bar{a}_{23} = \big(\bar{a}_{11} - \bar{a}_{33}\big)\big(\theta_x - \hat{\theta}_x\big)
$$

$$
\bar{a}_{13} = -\big(\bar{a}_{11} - \bar{a}_{33}\big)\cos\hat{\theta}_x\,\big(\theta_y - \hat{\theta}_y\big)
$$

$$
{}^{v}\bar{\mathbf{f}}_d[k] = {}^{v}\hat{\mathbf{A}}^{-1}[k]\left\{{}^{v}\bar{\mathbf{p}}_d[k+1] - {}^{v}\hat{\mathbf{p}}[k] - \lambda_c\left({}^{v}\bar{\mathbf{p}}_d[k] - {}^{v}\hat{\mathbf{p}}[k]\right)\right\}
$$

$$
{}^{v}\delta\bar{\mathbf{p}}[k+1] = \lambda_c\,{}^{v}\delta\bar{\mathbf{p}}[k] - \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right){}^{v}\bar{\mathbf{F}}_{dp}[k] - {}^{v}\boldsymbol{\varepsilon}_p[k]
$$

$$
\left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}\right){}^{v}\bar{\mathbf{F}}_{dp} = \mathbf{G}_a[k]\begin{bmatrix} \bar{a}_{11} - \hat{a}_{11} & \bar{a}_{33} - \hat{a}_{33} & \bar{a}_{13} - \hat{a}_{13} & \bar{a}_{23} - \hat{a}_{23} \end{bmatrix}^{T}
$$

$$
\mathbf{G}_a[k] = \begin{bmatrix} F_u & 0 & F_w & 0 \\ F_v & 0 & 0 & F_w \\ 0 & F_w & F_u & F_v \end{bmatrix}
$$

$$
{}^{v}\mathbf{M}[k] = {}^{v}\Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,{}^{v}\delta\bar{\mathbf{p}}_3[k]
$$

$$
{}^{v}\mathbf{w} = \begin{bmatrix} \cos\hat{\theta}_x\,(\theta_y - \hat{\theta}_y) & -(\theta_x - \hat{\theta}_x) & 1 \end{bmatrix}^{T}
$$

$$
\begin{aligned}
{}^{v}\delta\bar{\mathbf{p}}_1[k+1] &= {}^{v}\delta\bar{\mathbf{p}}_2[k] \\
{}^{v}\delta\bar{\mathbf{p}}_2[k+1] &= {}^{v}\delta\bar{\mathbf{p}}_3[k] \\
{}^{v}\delta\bar{\mathbf{p}}_3[k+1] &= \lambda_c\, {}^{v}\delta\bar{\mathbf{p}}_3[k] - \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right) {}^{v}\bar{\mathbf{F}}_{dp}[k] - {}^{v}\boldsymbol{\varepsilon}_p[k] \\
\bar{a}_{11}[k+1] &= \bar{a}_{11}[k] + \bar{a}'_{11}[k]\, {}^{v}\mathbf{w}^T \left({}^{v}\mathbf{M}[k] + \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right) {}^{v}\bar{\mathbf{F}}_{dp}[k] + {}^{v}\boldsymbol{\varepsilon}_p[k]\right) \\
\bar{a}_{33}[k+1] &= \bar{a}_{33}[k] + \bar{a}'_{33}[k]\, {}^{v}\mathbf{w}^T \left({}^{v}\mathbf{M}[k] + \left({}^{v}\bar{\mathbf{A}} - {}^{v}\hat{\mathbf{A}}[k]\right) {}^{v}\bar{\mathbf{F}}_{dp}[k] + {}^{v}\boldsymbol{\varepsilon}_p[k]\right) \\
\bar{a}'_{11}[k+1] &= \bar{a}'_{11}[k] \\
\bar{a}'_{33}[k+1] &= \bar{a}'_{33}[k] \\
\bar{a}_{13}[k+1] &= \bar{a}_{13}[k] \\
\bar{a}_{23}[k+1] &= \bar{a}_{23}[k] \\
\boldsymbol{\theta}[k+1] &= \boldsymbol{\theta}[k]
\end{aligned}
$$

3 Estimator

$$\begin{bmatrix} {}^v\delta\bar{\mathbf{p}}_1^T & {}^v\delta\bar{\mathbf{p}}_2^T & {}^v\delta\bar{\mathbf{p}}_3^T & \bar{a}_{11} & \bar{a}_{33} & \bar{a}'_{11} & \bar{a}'_{33} & \bar{a}_{13} & \bar{a}_{23} & \theta_x & \theta_y \end{bmatrix}^T \qquad (17 \times 1,\ d = 2)$$

$${}^v\delta\bar{\mathbf{p}}_m[k] = \hat{\mathbf{V}}^T[k]\,\delta\bar{\mathbf{p}}_m[k], \qquad {}^v\bar{\mathbf{A}}_m[k] = \hat{\mathbf{V}}^T[k]\,\bar{\mathbf{A}}_m[k]\,\hat{\mathbf{V}}[k]$$

$$\bar{a}_{m,11} = \tfrac{1}{2}\Big(\big[{}^v\bar{\mathbf{A}}_m\big]_{11} + \big[{}^v\bar{\mathbf{A}}_m\big]_{22}\Big), \qquad \bar{a}_{m,33} = \big[{}^v\bar{\mathbf{A}}_m\big]_{33}, \qquad \bar{a}_{m,13} = \big[{}^v\bar{\mathbf{A}}_m\big]_{13}, \qquad \bar{a}_{m,23} = \big[{}^v\bar{\mathbf{A}}_m\big]_{23}$$

$$\mathbf{y}_m[k] = \begin{bmatrix} {}^v\delta\bar{\mathbf{p}}_m[k] \\ \bar{a}_{m,11}[k] \\ \bar{a}_{m,33}[k] \\ \bar{a}_{m,13}[k] \\ \bar{a}_{m,23}[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} {}^v\delta\hat{\mathbf{p}}_1[k] \\ \hat{a}_{11}[k] \\ \hat{a}_{33}[k] \\ \hat{a}_{13}[k] \\ \hat{a}_{23}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_2 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_2 & \mathbf{0} \end{bmatrix}$$

$$\mathbf{e}_p[k] = {}^v\delta\bar{\mathbf{p}}_m[k] - {}^v\delta\hat{\mathbf{p}}_1[k]$$

$$\mathbf{e}_a[k] = \begin{bmatrix} \bar{a}_{m,11} - \hat{a}_{11} & \bar{a}_{m,33} - \hat{a}_{33} & \bar{a}_{m,13} - \hat{a}_{13} & \bar{a}_{m,23} - \hat{a}_{23} \end{bmatrix}^T$$

$$
\begin{aligned}
{}^{v}\hat{\mathbf{M}}[k] &= {}^{v}\Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,{}^{v}\delta\hat{\mathbf{p}}_3[k] \\
{}^{v}\delta\hat{\mathbf{p}}_1[k+1] &= {}^{v}\delta\hat{\mathbf{p}}_2[k] + \mathbf{L}_1[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
{}^{v}\delta\hat{\mathbf{p}}_2[k+1] &= {}^{v}\delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_2[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
{}^{v}\delta\hat{\mathbf{p}}_3[k+1] &= \lambda_c\,{}^{v}\delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_3[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
\hat{a}_{11}[k+1] &= \hat{a}_{11}[k] + \hat{a}'_{11}[k]\,\big[{}^{v}\hat{\mathbf{M}}[k]\big]_w + \mathbf{L}_4[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
\hat{a}_{33}[k+1] &= \hat{a}_{33}[k] + \hat{a}'_{33}[k]\,\big[{}^{v}\hat{\mathbf{M}}[k]\big]_w + \mathbf{L}_5[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
\hat{a}'_{11}[k+1] &= \hat{a}'_{11}[k] + \mathbf{L}_6[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
\hat{a}'_{33}[k+1] &= \hat{a}'_{33}[k] + \mathbf{L}_7[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
\hat{a}_{13}[k+1] &= \hat{a}_{13}[k] + \mathbf{L}_8[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
\hat{a}_{23}[k+1] &= \hat{a}_{23}[k] + \mathbf{L}_9[k]\big[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\big]^T \\
\hat{\boldsymbol{\theta}}[k+1] &= \hat{\boldsymbol{\theta}}[k] + \delta\hat{\theta}[k] \\
\delta\hat{\theta}_x[k] &= \frac{\hat{a}_{23}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]} \\
\delta\hat{\theta}_y[k] &= -\frac{\hat{a}_{13}[k+1]}{\big(\hat{a}_{11}[k] - \hat{a}_{33}[k]\big)\cos\hat{\theta}_x[k]}
\end{aligned}
$$

$$
{}^{v}\delta\hat{\mathbf{p}}_i \leftarrow \big(\hat{\mathbf{V}}^T[k]\,\hat{\mathbf{V}}[k+1]\big)^T\,{}^{v}\delta\hat{\mathbf{p}}_i
$$

$$
{}^{V[k+1]}\hat{\mathbf{A}}[k+1] \approx \mathrm{diag}\big(\hat{a}_{11}[k+1],\ \hat{a}_{22}[k+1],\ \hat{a}_{33}[k+1]\big), \qquad \hat{a}_{13}[k+1] \leftarrow 0, \qquad \hat{a}_{23}[k+1] \leftarrow 0
$$

4 Dynamics of estimation errors

$$\mathbf{e}[k] = \big[\, \big({}^v\delta\bar{\mathbf{p}}_1 - {}^v\delta\hat{\mathbf{p}}_1\big)^T \;\; \big({}^v\delta\bar{\mathbf{p}}_2 - {}^v\delta\hat{\mathbf{p}}_2\big)^T \;\; \big({}^v\delta\bar{\mathbf{p}}_3 - {}^v\delta\hat{\mathbf{p}}_3\big)^T$$
$$\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}'_{11} - \hat{a}'_{11} \;\; \bar{a}'_{33} - \hat{a}'_{33}$$
$$\bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23} \;\; \big(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\big)^T \,\big]^T$$

$$\theta_x - \hat{\theta}_x[k+1] = \big(\theta_x - \hat{\theta}_x[k]\big) - \frac{\hat{a}_{23}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]} = \frac{\bar{a}_{23}[k+1] - \hat{a}_{23}[k+1]}{\hat{a}_{11}[k] - \hat{a}_{33}[k]}$$
$$\theta_y - \hat{\theta}_y[k+1] = \big(\theta_y - \hat{\theta}_y[k]\big) + \frac{\hat{a}_{13}[k+1]}{\big(\hat{a}_{11}[k] - \hat{a}_{33}[k]\big)\cos\hat{\theta}_x[k]} = -\frac{\bar{a}_{13}[k+1] - \hat{a}_{13}[k+1]}{\big(\hat{a}_{11}[k] - \hat{a}_{33}[k]\big)\cos\hat{\theta}_x[k]}$$

$${}^v\delta\bar{\mathbf{p}}_3[k+1] - {}^v\delta\hat{\mathbf{p}}_3[k+1] = \lambda_c\big({}^v\delta\bar{\mathbf{p}}_3 - {}^v\delta\hat{\mathbf{p}}_3\big) - \mathbf{G}_a\big[\,\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23}\,\big]^T - {}^v\boldsymbol{\varepsilon}_p$$

$$\begin{aligned}
\bar{a}_{11}[k+1] - \hat{a}_{11}[k+1] &= \big(\bar{a}_{11} - \hat{a}_{11}\big) + \big(\bar{a}'_{11} - \hat{a}'_{11}\big)\big[{}^v\hat{\mathbf{M}}\big]_w + \hat{a}'_{11}(1-\lambda_c)\big[{}^v\delta\bar{\mathbf{p}}_3 - {}^v\delta\hat{\mathbf{p}}_3\big]_w \\
&\quad + \hat{a}'_{11}\Big[\mathbf{G}_a\big[\,\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23}\,\big]^T\Big]_w + \hat{a}'_{11}\big[{}^v\boldsymbol{\varepsilon}_p\big]_w \\
&\quad + \hat{a}'_{11}\Big(\cos\hat{\theta}_x\,\big[{}^v\hat{\mathbf{M}}\big]_u\big(\theta_y - \hat{\theta}_y\big) - \big[{}^v\hat{\mathbf{M}}\big]_v\big(\theta_x - \hat{\theta}_x\big)\Big) \\
\bar{a}_{33}[k+1] - \hat{a}_{33}[k+1] &= \big(\bar{a}_{33} - \hat{a}_{33}\big) + \big(\bar{a}'_{33} - \hat{a}'_{33}\big)\big[{}^v\hat{\mathbf{M}}\big]_w + \hat{a}'_{33}(1-\lambda_c)\big[{}^v\delta\bar{\mathbf{p}}_3 - {}^v\delta\hat{\mathbf{p}}_3\big]_w \\
&\quad + \hat{a}'_{33}\Big[\mathbf{G}_a\big[\,\bar{a}_{11} - \hat{a}_{11} \;\; \bar{a}_{33} - \hat{a}_{33} \;\; \bar{a}_{13} - \hat{a}_{13} \;\; \bar{a}_{23} - \hat{a}_{23}\,\big]^T\Big]_w + \hat{a}'_{33}\big[{}^v\boldsymbol{\varepsilon}_p\big]_w \\
&\quad + \hat{a}'_{33}\Big(\cos\hat{\theta}_x\,\big[{}^v\hat{\mathbf{M}}\big]_u\big(\theta_y - \hat{\theta}_y\big) - \big[{}^v\hat{\mathbf{M}}\big]_v\big(\theta_x - \hat{\theta}_x\big)\Big) \\
\bar{a}_{13}[k+1] - \hat{a}_{13}[k+1] &= \bar{a}_{13} - \hat{a}_{13} \\
\bar{a}_{23}[k+1] - \hat{a}_{23}[k+1] &= \bar{a}_{23} - \hat{a}_{23}
\end{aligned}$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] + \big[\,\mathbf{0} \;\; \mathbf{0} \;\; -{}^v\boldsymbol{\varepsilon}_p^T \;\; \hat{a}'_{11}[{}^v\boldsymbol{\varepsilon}_p]_w \;\; \hat{a}'_{33}[{}^v\boldsymbol{\varepsilon}_p]_w \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0\,\big]^T$$

5 Regulation

$$e_a[k] = \bar{a}_d - \hat{a}_{33}[k]$$

$$\Delta\bar{w}_d[k] = \frac{T_s}{\tau_P}\,\frac{e_a[k]}{\hat{a}'_{33}[k]} + \frac{T_s^2}{\tau_P\,\tau_I}\,\frac{\sum_{j\le k} e_a[j]}{\hat{a}'_{33}[k]}$$

$$\hat{a}_{33}[k+1] = \hat{a}_{33}[k] + \hat{a}'_{33}[k]\,\Delta\bar{w}_d[k]$$

$$e_a[k+1] = e_a[k] - \frac{T_s}{\tau_P}\,e_a[k] - \frac{T_s^2}{\tau_P\,\tau_I}\sum_{j\le k} e_a[j]$$

$$e_a[k+1] = \left(2 - \frac{T_s}{\tau_P} - \frac{T_s^2}{\tau_P\,\tau_I}\right)e_a[k] - \left(1 - \frac{T_s}{\tau_P}\right)e_a[k-1]$$

$$\frac{\hat{a}_{33}(s)}{\bar{a}_d(s)} = \frac{\dfrac{1}{\tau_P}\,s + \dfrac{1}{\tau_P\,\tau_I}}{s^2 + \dfrac{1}{\tau_P}\,s + \dfrac{1}{\tau_P\,\tau_I}}$$

$$s_z = -\frac{1}{\tau_I}$$

$$s^2 + \frac{1}{\tau_P}\,s + \frac{1}{\tau_P\,\tau_I} = 0$$

$$\frac{1}{\tau_P^2} = \frac{4}{\tau_P\,\tau_I}$$

$$\tau_I = 4\,\tau_P$$

6 Control force variance

$$f[k] = \frac{M[k]}{a_{nom}\,\bar{a}}$$

$$M[k] = (1-\lambda_c)\left(p_d - \hat{p}[k]\right)$$

$$p[k+1] = p[k] + M[k] + \varepsilon[k]$$

$$\hat{p}[k] = p[k] - \varepsilon[k-1] - \varepsilon[k-2] + n[k-2]$$

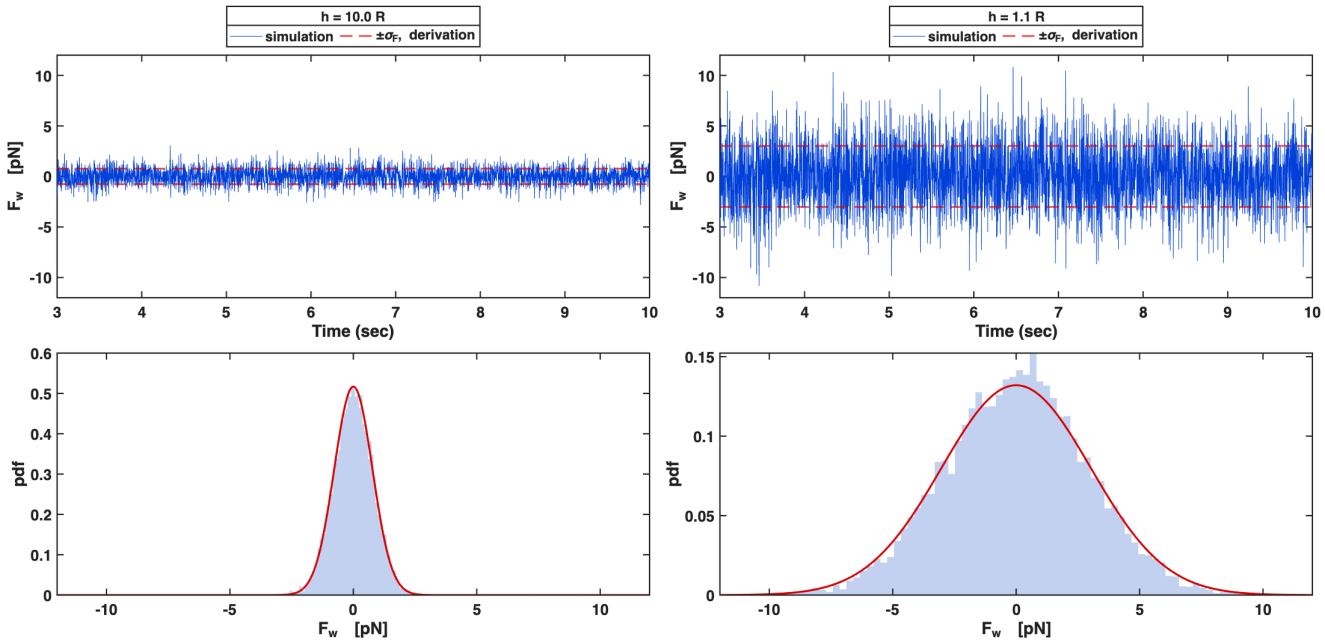
$$M(z) = -(1-\lambda_c)\,\frac{z^{-3}}{1-\lambda_c z^{-1}}\,\varepsilon(z) - (1-\lambda_c)\,\frac{z^{-2}\left(1-z^{-1}\right)}{1-\lambda_c z^{-1}}\,n(z)$$

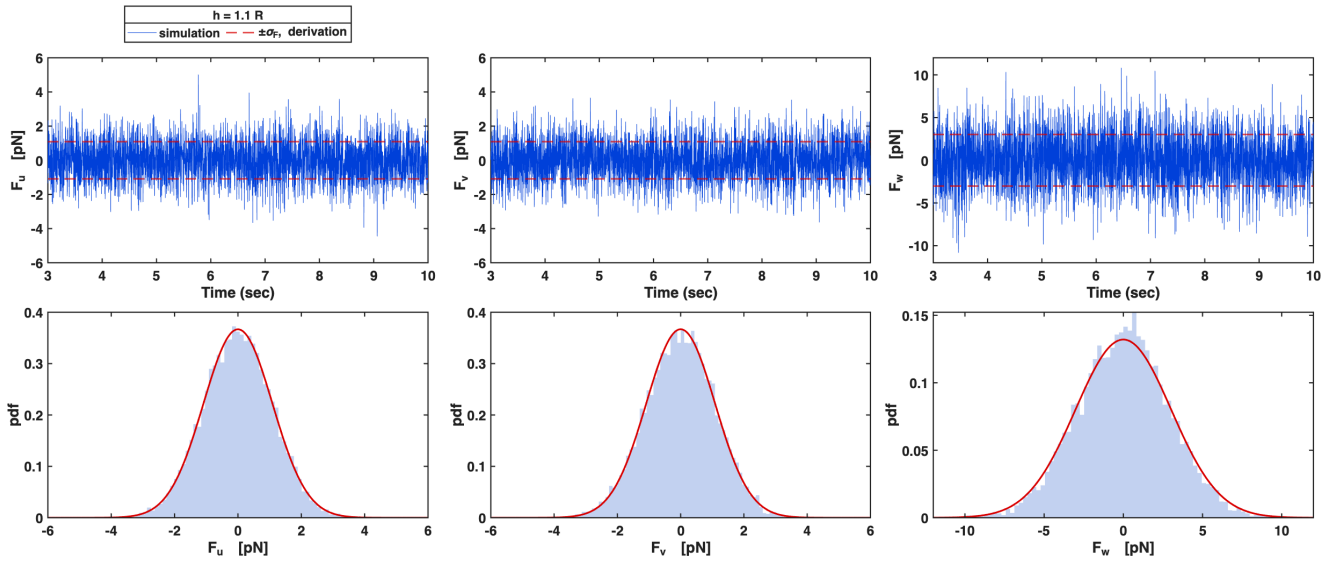
$$\sum_{j\geq 0} h_\varepsilon^2[j] = \frac{1}{1-\lambda_c^2}$$

$$\sum_{j\geq 0} h_n^2[j] = \frac{2}{1+\lambda_c}$$

$$\sigma_F^2 = \frac{1}{\left(a_{nom}\,\bar{a}\right)^2}\left(\frac{1-\lambda_c}{1+\lambda_c}\,\sigma_\varepsilon^2 + \frac{2\,(1-\lambda_c)^2}{1+\lambda_c}\,\sigma_n^2\right)$$

$$\sigma_\varepsilon^2 = 4\,k_B T\,a_{nom}\,\bar{a}$$


h = 10.0 R
simulation
±σF, derivation
Fw [pN]
Time (sec)
pdf
Fw [pN]
h = 1.1 R
simulation
±σF, derivation
Fw [pN]
Time (sec)
pdf
Fw [pN]


h = 1.1 R
simulation
±σF, derivation
Fu [pN]
Time (sec)
pdf
Fu [pN]
Fv [pN]
Time (sec)
pdf
Fv [pN]
Fw [pN]
Time (sec)
pdf
Fw [pN]